

Differential Geometry

Exercise 12

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12.1 Problem

Let G be a Lie group equipped with a bi-invariant metric. We show in this exercise that a 1-parameter subgroups and geodesics coincide. Let $\gamma: \mathbb{R} \rightarrow G$ be a geodesic starting at e .

- (1) Let B be a normal ball around e of radius $\varepsilon > 0$. Let s be small enough so that $\gamma|_{[0,s]} \subseteq B$. Let $\varphi = L_{\gamma(s)}$ be the isometry given by left-multiplication by $\gamma(s)$. Show that

$$\left. \frac{d}{dt} \right|_{t=0} (\varphi \circ \gamma) = \dot{\gamma}(s).$$

- (2) Conclude that $\varphi \circ \gamma(t) = \gamma(s+t)$ for all t . Thus for s sufficiently small and all t , $\gamma(s)\gamma(t) = \gamma(s+t)$.
 (3) Conclude $\gamma(s)\gamma(t) = \gamma(s+t)$ for all s, t .

- (1) Since $\gamma|_{[0,s]}$ is contained in a normal ball, it is a radial geodesic and thus minimizing. Let $s' \in (0, s)$, then consider the curve α obtained by concatenating $\gamma|_{[s',s]}$ and $\varphi \circ \gamma|_{[0,s']}$. This is a curve $\gamma(s')$ to $\varphi \circ \gamma(s')$.

Furthermore, since $\gamma|_{[s',s]}$ is minimizing and φ is an isometry, so is $\varphi \circ \gamma|_{[0,s']}$. Thus

$$L(\alpha) = L(\gamma|_{[s',s]}) + L(\varphi \circ \gamma|_{[0,s']}) = d(\gamma(s'), \gamma(s)) + d(\varphi(e), \varphi \circ \gamma(s')) = d(\gamma(s'), \gamma(s)) + d(e, \gamma(s')).$$

Since all these points, $e = \gamma(0), \gamma(s'), \gamma(s)$, lie on the minimizing γ this is equal to $d(e, \gamma(s))$. Since right-multiplication by $\gamma(s')$ is also an isometry,

$$L(\alpha) = d(e, \gamma(s)) = d(e \cdot \gamma(s'), \gamma(s) \cdot \gamma(s')) = d(\gamma(s'), \varphi \circ \gamma(s')).$$

Thus α is minimizing and thus a geodesic. Since $\alpha(t) = \gamma(t+s')$ for $t < s-s'$, by the uniqueness of geodesics we must have $\alpha(t) = \gamma(t+s')$ for all t . In particular

$$\gamma(s+t) = \alpha(s-s'+t) = \varphi \circ \gamma(t),$$

and thus differentiating both sides and evaluating at $t = 0$ gives

$$\left. \frac{d}{dt} \right|_{t=0} \varphi \circ \gamma(t) = \dot{\gamma}(s)$$

as desired.

- (2) As φ is an isometry, $\varphi \circ \gamma$ is a geodesic starting at $\varphi(e) = \gamma(s)$ with initial velocity $\dot{\gamma}(s)$. By uniqueness of geodesics, $\varphi \circ \gamma(t) = \gamma(t+s)$ for all t , since $\gamma(t+s)$ is another geodesic with these same initial conditions. By definition of φ this means $\gamma(s)\gamma(t) = \gamma(t+s)$ for all t .
 (3) Let s be arbitrary, then there is some natural n such that s/n is small enough to apply the above point (i.e. $\gamma|_{[0,s/n]}$ is contained in B). Then for all t we have $\gamma(s/n+t) = \gamma(s/n)\gamma(t)$.

In particular,

$$\gamma(s+t) = \gamma\left(n\frac{s}{n}+t\right) = \gamma\left(\frac{s}{n}\right)\gamma\left((n-1)\frac{s}{n}+t\right),$$

and continuing this inductively,

$$\gamma(s+t) = \gamma\left(\frac{s}{n}\right)^n \gamma(t).$$

In particular for $t = 0$ we see that $\gamma(s) = \gamma(s/n)^n$, and so for arbitrary t we have

$$\gamma(s+t) = \gamma\left(\frac{s}{n}\right)^n \gamma(t) = \gamma(s)\gamma(t)$$

as desired.

12.2 Problem

Let M be an orientable Riemannian n -manifold with positive curvature and even dimension. Let γ be a closed geodesic in M , i.e. an immersion $S^1 \rightarrow M$ which is a geodesic at all points. Prove that γ is homotopic to a closed curve whose length is strictly less than γ 's.

Let $w = \dot{\gamma}(0)$, where 0 denotes any distinguished point on S^1 , we will also view γ as a smooth geodesic $[0, 1] \rightarrow M$. Let $e_1, \dots, e_{n-1}$ be an orthonormal basis for the orthogonal complement $w^\perp$. Now consider parallel transport P^γ ; we claim there is a tangent vector $v \in w^\perp$ such that $P^\gamma v = v$.

Firstly, P^γ is an isometry because the connection is metric (since $\langle V, W \rangle$ is constant when V, W are parallel along γ). Thus $P^\gamma(e_i)$ forms an orthonormal basis, so there is an orthogonal matrix A such that $P^\gamma(e_i) = A_i^j e_j$. Furthermore we know that $\dot{\gamma}$ is parallel along γ as it is a geodesic, so $P^\gamma w = \dot{\gamma}(1) = \dot{\gamma}(0) = w$.

Let ω be an orientation form for M . Let V_i be the parallel transports of e_i , and $W = \dot{\gamma}$ is w 's; then $\omega(V_1(t), \dots, V_{n-1}(t), W(t)) \neq 0$ for all t as $V_1(t), \dots, V_{n-1}(t), W(t)$ forms a basis for $T_{\gamma(t)}M$ as parallel transport is a linear isomorphism. So $\omega(\bar{V}(t), W(t))$ is a continuous non-vanishing map, since its domain is connected this means $\omega(\bar{V}(1), W(1)) = \omega(\bar{V}(1), w)$ and $\omega(\bar{V}(0), W(0)) = \omega(e_1, \dots, e_{n-1}, w)$ have the same sign.

Since

$$\omega(\bar{V}(1), w) = \omega(A\bar{e}, w) = \det(A \oplus (1))\omega(\bar{e}, w) = \det A \omega(\bar{e}, w),$$

this means that $\det A > 0$. Thus A is an orthogonal matrix with positive determinant of odd dimension $(n-1)$; we showed in class that this means that it has a nonzero fixed point $(a^1, \dots, a^{n-1})$. Let $v = a^i e_i$, so that

$$P^\gamma(v) = a^i P^\gamma(e_i) = A_i^j a^i e_j = a^j e_j = v.$$

Let $V(t)$ be v 's parallel transport along γ . Since $V(1) = v = V(0)$, this means we can view V as a smooth vector field $S^1 \rightarrow M$ along γ . Let Γ be a proper variation of γ with variation field V , and let $E(s) = E(\Gamma_s)$. Since γ is a geodesic $E'(0) = 0$. Furthermore since V is smooth and proper, and Γ is smooth and proper as well (or at least can be taken to be so),

$$\frac{1}{2}E''(0) = - \int_0^1 \langle V(t), V''(t) + R(\dot{\gamma}(t), V(t))\dot{\gamma}(t) \rangle dt.$$

Since V is parallel along γ , $V' = 0$ and thus $V'' = 0$. So this is equal to

$$\frac{1}{2}E''(0) = - \int_0^1 \langle V(t), R(\dot{\gamma}(t), V(t))\dot{\gamma}(t) \rangle dt = - \int_0^1 \text{Rm}(\dot{\gamma}(t), V(t), \dot{\gamma}(t), V(t)) dt.$$

The integrand is a positive multiple of the sectional curvature (since $v = V(0)$ and $w = \dot{\gamma}(0)$ are linearly independent, $V(t), \dot{\gamma}(t)$ remain linearly independent), so this integral is positive. Thus $E''(0)$ is negative, and since $E'(0) = 0$, this means 0 is a local maximum of $E(s)$.

Let $s > 0$ be such that $E(s) < E(0)$. Then $\gamma_s(t) = \Gamma(s, t)$ is a smooth curve where

$$L(\gamma_s) \leq E(\gamma_s)^2 = E(s)^2 < E(0)^2 = E(\gamma)^2 = L(\gamma),$$

where the last equality is because γ is a geodesic defined on $[0, 1]$. And γ, γ_s are homotopic, as Γ forms a homotopy between them (since by definition $\gamma(t) = \Gamma(0, t)$ and $\gamma_s(t) = \Gamma(s, t)$).

12.3 Problem

Let $f: \bar{M} \rightarrow M$ be a Riemannian submersion, meaning it is a surjective submersion and it preserves the metric between vectors orthogonal to its fibers.

A vector $\bar{x} \in T_{\bar{p}}\bar{M}$ is **horizontal** if it is orthogonal to the fiber $f^{-1}p$ ($p = f\bar{p}$). Similarly it is **vertical** if it is tangent to the fiber. Thus there is a decomposition,

$$T_{\bar{p}}\bar{M} = (T_{\bar{p}}\bar{M})^h \oplus (T_{\bar{p}}\bar{M})^v, \quad (T_{\bar{p}}\bar{M})^v = T_{\bar{p}}f^{-1}p, \quad (T_{\bar{p}}\bar{M})^h = (T_{\bar{p}}\bar{M}^v)^\perp.$$

For $X \in \mathfrak{X}(M)$, the **horizontal lift** of $\bar{X}$ of X is the horizontal vector field defined by $df_{\bar{p}}(\bar{X}|_{\bar{p}}) = X|_p$.

(1) Show that $\bar{X}$ is differentiable.

(2) Let $\nabla, \bar{\nabla}$ be the Riemannian connections of $M, \bar{M}$ respectively. Show that

$$\bar{\nabla}_{\bar{X}}\bar{Y} = \bar{\nabla}_X\bar{Y} + \frac{1}{2}[\bar{X}, \bar{Y}]^v, \quad X, Y \in \mathfrak{X}(M),$$

where Z^v is the vertical component of $Z \in \mathfrak{X}(\bar{M})$.

(3) Show that $[\bar{X}, \bar{Y}]_{\bar{p}}$ depends only on $\bar{X}|_{\bar{p}}$ and $\bar{Y}|_{\bar{p}}$.

- (1) Notice that $df_{\bar{p}}$ is an isomorphism $T_{\bar{p}}\bar{M}^h \rightarrow T_pM$, as f is constant on its fibers and thus df_p is zero on $T_{\bar{p}}\bar{M}^v$. Since df_p is surjective as f is a submersion, and the dimensions of the horizontal part and T_pM are equal, this means that this must be an isomorphism. Thus $\bar{X}$ is uniquely determined; we need only show it is smooth.

Since f is a submersion, there are local charts $\bar{p} \in (U, (\bar{x}, \bar{y}))$ and $p \in (V, (\tilde{x}))$ such that f has the local representation $(\bar{x}, \bar{y}) \mapsto \tilde{x}$. Thus $df(\partial/\partial \tilde{x}^i) = \partial/\partial \tilde{x}^i \circ f$ and $df(\partial/\partial y^j) = 0$. This means that the $\partial/\partial y^j$ span $T_{\bar{p}}\bar{M}^v$. Applying Gram-Schmidt to $\partial/\partial y^j, \partial/\partial \tilde{x}^i$ (in this order) gives a smooth local orthonormal frame $\bar{F}, \bar{E}$. The span of $\bar{F}$ and $\partial/\partial y^j$ are the same, so $\bar{F}$ is an orthonormal frame of $T_{\bar{p}}\bar{M}^v$ and thus $\bar{E}$ is an orthonormal frame of $T_{\bar{p}}\bar{M}^h$.

Now, there are smooth $a^i \in C^\infty(V)$ such that locally $X = a^i \partial/\partial \tilde{x}^i$. Define $\bar{Y} = (a^i \circ f) \partial/\partial \tilde{x}^i$, which is a local vector field of $\bar{M}$ and

$$df\bar{Y} = (a^i \circ f) \frac{\partial}{\partial \tilde{x}^i} \circ f = X \circ f.$$

We can project $\bar{Y}$ onto its $E_1, \dots, E_n$ components, which is smooth since F_j, E_i form a local frame. Since the other components F_j are in df 's kernel, this does not affect the image. That is to say, $\bar{Y}^h$ is smooth and $df\bar{Y}^h = df\bar{Y} = X \circ f$.

By uniqueness, $\bar{Y}^h = \bar{X}$ in this neighborhood. Since $\bar{Y}^h$ is smooth, so too must $\bar{X}$ be in this neighborhood. This neighborhood was chosen around an arbitrary point, so $\bar{X}$ is smooth.

- (2) Let $X, Y, Z \in \mathfrak{X}(M)$ and $T \in \mathfrak{X}(\bar{M})$ be a vertical field. Notice that every vector field of $\mathfrak{X}(\bar{M})$ has a horizontal and vertical component. By the same argument as the previous point, the horizontal and vertical components of a vector field are smooth. This is because we can find a local orthonormal frame $\bar{F}, \bar{E}$ for the vertical and horizontal tangent bundles and then the components of a vector field in $\mathfrak{X}(\bar{M})$ are just the projections onto $\bar{F}$ and $\bar{E}$ respectively, which is smooth.

Anyway, since $\bar{X}$ is horizontal and T is vertical, $\langle \bar{X}, T \rangle = 0$ (at every point they are by definition orthogonal). Since $\bar{Y}, \bar{Z}$ are horizontal, and f is Riemannian and thus preserves the metric between them,

$$\bar{X}\langle \bar{Y}, \bar{Z} \rangle = \bar{X}\langle df\bar{Y}, df\bar{Z} \rangle = \bar{X}(\langle Y, Z \rangle \circ f) = df(\bar{X})\langle Y, Z \rangle = X\langle Y, Z \rangle.$$

The second-to-last equality is because $df(v)(u) = v(u \circ f)$ for all smooth real-valued u and derivations v . Similarly,

$$T\langle \bar{X}, \bar{Y} \rangle = T(\langle X, Y \rangle \circ f) = df(T)\langle X, Y \rangle = 0$$

for the same reason.

We also have $df[\bar{X}, T] = [df\bar{X}, df(T)] = [X, 0] = 0$. And $df[\bar{X}, \bar{Y}] = [df\bar{X}, df\bar{Y}] = [X, Y]$.

Notice that for $V \in \mathfrak{X}(\bar{M})$ arbitrary, since $\bar{X}$ is horizontal

$$\langle V, \bar{X} \rangle = \langle V^h, \bar{X} \rangle = \langle df(V^h), df\bar{X} \rangle = \langle df(V), X \rangle.$$

This is because projecting onto the horizontal component is an orthogonal projection, and $\bar{X}$ is already horizontal; and $df(V) = df(V^h) + df(V^v) = df(V^h)$ since df is zero on vertical vectors. Thus we have

$$\langle [\bar{X}, \bar{Y}], \bar{Z} \rangle = \langle df[\bar{X}, \bar{Y}], df\bar{Z} \rangle = \langle [X, Y], Z \rangle,$$

and

$$\langle [\bar{X}, T], \bar{Y} \rangle = \langle df[\bar{X}, T], df\bar{Y} \rangle = 0.$$

By Koszul's formula,

$$\langle \bar{\nabla}_{\bar{X}} \bar{Y}, \bar{Z} \rangle = \frac{1}{2} (\bar{X} \langle \bar{Y}, \bar{Z} \rangle + \bar{Y} \langle \bar{Z}, \bar{X} \rangle - \bar{Z} \langle \bar{X}, \bar{Y} \rangle - \langle \bar{Y}, [\bar{X}, \bar{Z}] \rangle - \langle \bar{Z}, [\bar{Y}, \bar{X}] \rangle + \langle \bar{X}, [\bar{Z}, \bar{Y}] \rangle).$$

Our computations showed that this is equal to the same expression but with X, Y, Z instead of $\bar{X}, \bar{Y}, \bar{Z}$. By Koszul's formula, this is just $\langle \nabla_X Y, Z \rangle$ so

$$\langle \bar{\nabla}_{\bar{X}} \bar{Y}, \bar{Z} \rangle = \langle \nabla_X Y, Z \rangle.$$

Again by Koszul,

$$\langle \bar{\nabla}_{\bar{X}} \bar{Y}, T \rangle = \frac{1}{2} (\bar{X} \langle \bar{Y}, T \rangle + \bar{Y} \langle T, \bar{X} \rangle - T \langle \bar{X}, \bar{Y} \rangle - \langle \bar{Y}, [\bar{X}, T] \rangle - \langle T, [\bar{Y}, \bar{X}] \rangle + \langle \bar{X}, [T, \bar{Y}] \rangle).$$

As we showed, all of these terms are zero except for $-\langle T, [\bar{Y}, \bar{X}] \rangle = \langle [\bar{X}, \bar{Y}], T \rangle$. Since T is vertical, this is equal to $\langle [\bar{X}, \bar{Y}]^v, T \rangle$. Thus

$$\langle \bar{\nabla}_{\bar{X}} \bar{Y}, T \rangle = \frac{1}{2} \langle [\bar{X}, \bar{Y}]^v, T \rangle.$$

And so

$$\langle \bar{\nabla}_{\bar{X}} \bar{Y}, \bar{Z} + T \rangle = \langle \nabla_X Y, Z \rangle + \left\langle \frac{1}{2} [\bar{X}, \bar{Y}]^v, T \right\rangle.$$

Since $\langle X, Z \rangle = \langle \bar{X}, \bar{Z} \rangle$, we see (also because $\bar{Z}$ is horizontal and thus orthogonal to vertical vectors, and T is orthogonal to horizontal vectors),

$$= \left\langle \bar{\nabla}_X \bar{Y} + \frac{1}{2} [\bar{X}, \bar{Y}]^v, \bar{Z} + T \right\rangle.$$

If we evaluate at a specific point $\bar{p}$ where all tangent vectors can be decomposed to their vertical and horizontal components, and the horizontal component is the value of a horizontal lift of some vector field of M , we get that at $\bar{p}$ for all $v \in T_{\bar{p}} \bar{M}$

$$\langle \bar{\nabla}_{\bar{X}} \bar{Y}|_{\bar{p}}, v \rangle = \left\langle \bar{\nabla}_X \bar{Y}|_{\bar{p}} + \frac{1}{2} [\bar{X}, \bar{Y}]_{\bar{p}}^v, v \right\rangle,$$

and thus we have equality

$$\overline{\nabla}_X \overline{Y} = \overline{\nabla_X Y} + \frac{1}{2}[\overline{X}, \overline{Y}]^v$$

at $\overline{p}$, and thus over all of $\overline{M}$ as $\overline{p}$ is arbitrary.

More explicitly, for tangent v , write $v = v^v + v^h$ and then let $Z \in \mathfrak{X}(M)$ be such that $Z_p = df_p(v^h)$ so that $\overline{Z}_{\overline{p}} = v^h$. And extend v^v to a vertical vector field T . Then we get the desired result.

- (3) By additivity, it is sufficient to show that if $\overline{X}(\overline{p}) = 0$ then $[\overline{X}, \overline{Y}]_{\overline{p}}^v = 0$ (and similarly for $\overline{Y}$, but this is symmetrical). To show that $[\overline{X}, \overline{Y}]^v = 0$ it is sufficient to show that $\langle [\overline{X}, \overline{Y}]^v, V \rangle = 0$ for arbitrary $V \in \mathfrak{X}(\overline{M})$. For horizontal V this is clearly true, so it is sufficient to show this for $V = T$ vertical in which case

$$\langle [\overline{X}, \overline{Y}]^v, T \rangle = \langle [\overline{X}, \overline{Y}], T \rangle = \langle \overline{\nabla}_X \overline{Y} - \overline{\nabla}_Y \overline{X}, T \rangle.$$

(Everything here is being evaluated at $\overline{p}$).

Since $\overline{\nabla} \overline{Y}$ is tensorial, since $\overline{X}_{\overline{p}} = 0$ we have $\overline{\nabla}_X \overline{Y} = 0$ at $\overline{p}$. So we just need to show that $\overline{\nabla}_Y \overline{X}$ is horizontal. Let (E_i) be a local horizontal frame around $\overline{p}$, so that $\overline{X} = u^i E_i$ for smooth real-valued u^i (since $\overline{X}$ is horizontal it has a representation by E_i). Since $X|_{\overline{p}} = 0$ this means $u^i(\overline{p}) = 0$ for all i . Then

$$\overline{\nabla}_Y \overline{X} = \overline{\nabla}_Y u^i E_i = u^i \overline{\nabla}_Y E_i + (\overline{Y} u^i) E_i.$$

Evaluating at $\overline{p}$, the first term vanishes as $u^i(\overline{p}) = 0$. The second term is horizontal as the E_i are. Thus at $\overline{p}$, $\overline{\nabla}_Y \overline{X}$ is horizontal and thus orthogonal to T . So $[\overline{X}, \overline{Y}]^v$ is orthogonal to all vertical T , and thus is zero.

Thus we are done, but I spell out the end. Firstly, clearly $\overline{X'} - \overline{X} = \overline{X} - \overline{X'}$ as both are horizontal lifts of $X - X'$ by linearity of df ; also projection $\bullet^v$ is linear. If $\overline{X}_{\overline{p}}' = \overline{X}_{\overline{p}}$ then at $\overline{p}$, $[\overline{X}, \overline{Y}]^v - [\overline{X'}, \overline{Y}]^v = [\overline{X} - \overline{X'}, \overline{Y}]^v = 0$ as we just showed. Thus $[\overline{X}, \overline{Y}]^v = [\overline{X'}, \overline{Y}]^v$ at $\overline{p}$.

For $\overline{Y}$, if $\overline{Y}_{\overline{p}} = 0$ then $[\overline{Y}, \overline{X}]^v = 0$ at $\overline{p}$ and thus $[\overline{X}, \overline{Y}]^v = -[\overline{Y}, \overline{X}]^v = 0$. Thus we obtain the result for $\overline{Y}$. So $[\overline{X}, \overline{Y}]^v$'s value at $\overline{p}$ is determined only by $\overline{X}$ and $\overline{Y}$'s.

12.4 Problem

For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, let $E \rightarrow M$ be a smooth $\mathbb{K}$ -vector bundle of rank r over a smooth manifold M .

- (1) Consider the r -fold fiber product $E^r = E \times_M \cdots \times_M E \rightarrow M$, which is a smooth manifold. By construction, its fibers over $p \in M$ are $E_p \oplus \cdots \oplus E_p$. Let $\text{Fr}(E) \subseteq E^r$ be the subset given by tuples $v_1, \dots, v_r \in E_p$ of bases, as p ranges. Show that $\text{Fr}(E)$ is open and thus a submanifold.
- (2) Show that $\text{GL}_r(\mathbb{K})$ acts properly, smoothly, and freely on $\text{Fr}(E)$ by $(a_i^j) \cdot (p, (v_i)) = (p, (a_i^j v_j))$. In the case that $\mathbb{K} = \mathbb{R}$, show that this action restricts to a proper, smooth, free action by the subgroup $\text{GL}_r^+(\mathbb{R})$ of matrices of positive determinant.
- (3) Show that $\text{Fr}(E)/\text{GL}_r(\mathbb{K})$ is diffeomorphic to M . When $\mathbb{K} = \mathbb{R}$, define the quotient $\text{Or}(E) = \text{Fr}(E)/\text{GL}_r^+(\mathbb{R})$. Show that $\text{Or}(E) \rightarrow M$ is a two-fold cover of M .
We say that E is **orientable** if there is a section of the covering $\text{Or}(E) \rightarrow M$. An **orientation** is a choice of a section.
- (4) Show that M is orientable (as a smooth manifold) iff TM is (as a vector bundle).
- (5) The Mobius band M is the quotient of $S^1 \times \mathbb{R}$ by the action of $\mathbb{Z}/2$ where the action by the non-trivial element is given by $(\theta, x) \mapsto (\theta + \pi, -x)$. Show that M is not orientable by showing that TM is not.
- (6) Let $\pi: \text{Or}(E) \rightarrow M$ be the two-fold cover of M from point (3). Show that the pullback bundle $\pi^* E \rightarrow \text{Or}(E)$ has a natural orientation. Deduce that $\text{Or}(TM)$ is orientable as a manifold.

- (1) Let $\pi_E: E \rightarrow M$ be the projection map. Let $\Phi: \pi_E^{-1}U \rightarrow U \times \mathbb{K}^r$ be a local trivialization, and define its r -fold product $\Phi^{\times r}: (\pi_E^{-1}U)^{\times r} \rightarrow U^{\times r} \times \mathbb{K}^{r \times r}$. Since E^r is an embedded submanifold of $E^{\times r}$ (as shown during the construction of the fiber product), restricting $\Phi^{\times r}$ to E^r remains smooth; call this restriction Φ^r . In fact as the fiber product of vector bundles, Φ^r is a local trivialization of E^r . So we have in particular $(\pi_E^{-1}U)^{\times r} = \pi_{E^r}^{-1}U$.

Now consider the composition

$$\tilde{\Phi}^r: \pi_{E^r}^{-1}U \xrightarrow{\Phi^r} U^{\times r} \times \mathbb{K}^{r \times r} \xrightarrow{\text{proj}} \mathbb{K}^{r \times r} \xrightarrow{\det} \mathbb{K}.$$

This maps a tuple $(v_1, \dots, v_r)$ in E^r to their representation by the local frame associated to Φ , and then to the determinant of this matrix. This tuple is linearly independent iff the representation by the local frame is, which is iff the representation matrix is invertible. Thus

$$\text{Fr}(E) \cap (\pi_{E^r}^{-1}U = (\tilde{\Phi}^r)^{-1}(\mathbb{K} - 0),$$

which is open as $\tilde{\Phi}^r$ is continuous. Since the trivializations cover E^r , this means that $\text{Fr}(E)$ is open as desired.

- (2) This is clearly a group action. We consider the representation of the action in each local trivialization of E^r , under which $\text{Fr}(E)$ has the image of $\text{GL}_r(\mathbb{K})$. In each local trivialization, this group action corresponds to multiplication $\text{GL}_r(\mathbb{K}) \times \text{GL}_r(\mathbb{K}) \rightarrow \text{GL}_r(\mathbb{K})$ (this follows from the linearity of the trivializations on the fibers). This is smooth in each local trivialization, and thus smooth on all of $\text{Fr}(E)$. The action of $\text{GL}_r(\mathbb{K})$ on itself is free and proper as well, and thus so is this action.

Since the action of $\text{GL}_r(\mathbb{R})$ is free and smooth, so is $\text{GL}_r^+(\mathbb{R})$'s as an open subgroup. And it is proper since $\text{GL}_r^+(\mathbb{R})$ is a connected component of $\text{GL}_r(\mathbb{R})$ and thus closed in $\text{GL}_r(\mathbb{R})$. In general if $H \subseteq G$ is a closed subgroup and G acts properly on M , so does H : for $K \subseteq M$ compact, $H_K = G_K \cap H$. Since G_K is compact and H is closed, H_K is a closed subset of G_K and thus compact.

- (3) The projection $\bar{\pi}: \text{Fr}(E)/\text{GL}_r(\mathbb{K}) \rightarrow M$ is bijective. This is because we can construct an inverse by mapping $p \in M$ to $(p, [e_1, \dots, e_r])$ where $e_1, \dots, e_r$ is any basis of E_p . This is independent on choice of basis, as they can be mapped to one another via a transition matrix.

So we have the commutative diagram

$$\begin{array}{ccc} \text{Fr}(E) & & \\ \pi_{\text{GL}} \downarrow & \searrow \pi_{\text{Fr}} & \\ \text{Fr}(E)/\text{GL}_r(\mathbb{K}) & \xrightarrow{\bar{\pi}} & M \end{array}$$

The maps π_{GL} and π_{Fr} are submersions (the first because $E^r \rightarrow M$ is a vector bundle and $\text{Fr}(E) \subseteq E^r$, the second because quotient maps are). So

$$\bar{\pi} \circ \pi_{\text{GL}} = \pi_{\text{Fr}},$$

and thus $\bar{\pi}$ is smooth (because π_{GL} is a submersion, $\bar{\pi}$ is smooth iff $\bar{\pi} \circ \pi_{E^r}$ is). And similarly

$$\bar{\pi}^{-1} \circ \pi_{\text{Fr}} = \pi_{\text{GL}}$$

means that $\bar{\pi}^{-1}$ is smooth as well. Thus $\bar{\pi}$ is a diffeomorphism $\text{Fr}(E)/\text{GL}_r(\mathbb{K}) \cong M$.

Now consider projection $\pi: \text{Or}(E) \rightarrow M$, so we have the commutative diagram

$$\begin{array}{ccc} \text{Fr}(E) & & \\ \pi_{\text{Or}} \downarrow & \searrow \pi_{\text{Fr}} & \\ \text{Or}(E) & \xrightarrow{\pi} & M \end{array}$$

For the same reason as before, π is then smooth.

If we take a local trivialization Φ^r of $U \subseteq M$, then $\pi^{-1}U = \pi_{\text{Or}}(\pi_{\text{Fr}}^{-1}(U))$. Under the trivialization Φ^r , $\pi_{\text{Fr}}^{-1}U$ corresponds to $U \times \text{GL}_r(\mathbb{R}) = (U \times \text{GL}_r^+(\mathbb{R})) \sqcup (U \times \text{GL}_r^-(\mathbb{R}))$. And thus

$$\pi^{-1}U = \pi_{\text{Or}}(U \times \text{GL}_r^+(\mathbb{R})) \sqcup \pi_{\text{Or}}(U \times \text{GL}_r^-(\mathbb{R})),$$

the union is disjoint because $\text{GL}_r^+(\mathbb{R})$'s action preserves the sign of the determinant.

π is bijective on both of these components: its inverse for the first is given by mapping $p \in U$ to $(p, [e_1, \dots, e_r])$ where $e_1, \dots, e_r$ is the local frame associated with Φ^r . The inverse for the second is given by mapping $p \mapsto (p, [-e_1, e_2, \dots, e_r])$.

Both of these are smooth for the same reason as above: calling them π_1^{-1}, π_2^{-1} we have $\pi_{\text{Or}} = \pi_i^{-1} \circ \pi_{\text{Fr}}$, and since $\pi_{\text{Or}}, \pi_{\text{Fr}}$ are submersions this implies smoothness. Thus π is indeed a two-fold cover of M as desired.

- (4) If M is orientable, let $(U_i, (x_i^j))$ be a consistently oriented atlas for M . Define $\psi_i: U_i \rightarrow \text{Fr}(TM)$ by $\psi_i(p) = \left(\partial/\partial x_i^1|_p, \dots, \partial/\partial x_i^n|_p \right)$. This is smooth as $\text{Fr}(TM)$ is a submanifold of $TM^{\times r}$, and this is a local section of TM^r (if we take the local trivialization Φ associated with $\partial/\partial x_i^j|_p$, this map under Φ^r corresponds to $p \mapsto (p, \text{id})$ which is smooth). Thus this defines a map to the quotient $\bar{\psi}_i: U_i \rightarrow \text{Or}(TM)$ which is a local section.

Since the atlases are consistently oriented, $\psi_j(p) = \left(\partial/\partial x_j^1|_p, \dots, \partial/\partial x_j^n|_p \right)$'s transition matrix with $\psi_i(p)$ has positive determinant. Thus $\pi_{\text{Or}}(\psi_j(p)) = \pi_{\text{Or}}(\psi_i(p))$, which means $\bar{\psi}_i = \bar{\psi}_j$ along their intersections. So we can glue these local sections together to get a global section, so $\text{Or}(TM)$ is oriented as desired.

Conversely let $\sigma: M \rightarrow \text{Or}(TM)$ be a global section of $\text{Or}(TM) \rightarrow M$. Then for each $p \in M$, given $T_p M$ the orientation determined by $\sigma(p)$. That is, $e_1, \dots, e_n \in T_p M$ is positively oriented iff $[e_1, \dots, e_n] \in \sigma(p)$.

We claim that this is a continuous orientation. For $p \in M$ take a connected local chart $p \in (U, (x^i))$. If $[\partial/\partial x^1|_p, \dots, \partial/\partial x^n|_p] \notin \sigma(p)$ then since the fibers of $\text{Or}(TM)$ have cardinality 2 this means $[-\partial/\partial x^1|_p, \dots, \partial/\partial x^n|_p] \in \sigma(p)$, so we can swap x^1 with $-x^1$ to obtain a chart whose coordinates are in $\sigma(p)$. Then define $\hat{\sigma}: U \rightarrow \text{Fr}(E)$ by $\hat{\sigma}(q) = (\partial/\partial x^1|_q, \dots, \partial/\partial x^n|_q)$, so we have that $\pi_{\text{Or}} \circ \hat{\sigma}(p) = \sigma(p)$; we claim this is true for all $q \in U$.

Now consider $\text{sgn}: \text{Fr}(TM) \rightarrow \{\pm 1\}$ which maps $(e_1, \dots, e_n) \in E_q$ to $+1$ if $(e_1, \dots, e_n)$ and $\hat{\sigma}(q)$ are consistently oriented, and -1 otherwise. This just maps $(e_1, \dots, e_n)$ to the sign of the determinant of its transition matrix with $\hat{\sigma}(q)$. This is thus continuous, as taking the determinant and sign are.

Since sgn is invariant under $\text{GL}_r^+(\mathbb{R})$'s action, it quotients to give a continuous map which we also denote by $\text{sgn}: \text{Or}(TM) \rightarrow \{\pm 1\}$. This is just the indicator for whether or not $\hat{\sigma}$ is an orientation: $\text{sgn}(\mathcal{O}_q) = +1$ if $\hat{\sigma}(q) \in \mathcal{O}_q$ and -1 otherwise. Now consider $\text{sgn} \circ \sigma: U \rightarrow \{\pm 1\}$: it maps q to $+1$ if $\hat{\sigma}(q) \in \sigma(q)$ and -1 otherwise. As a composition of continuous maps, it is continuous; since $\{\pm 1\}$ is discrete it is locally constant. Since U is connected it is constant, and since $\hat{\sigma}(p) \in \sigma(p)$ it is identically 1.

Thus $\partial/\partial x^1, \dots, \partial/\partial x^n$ are in σ for all points in U , meaning it is an oriented local coordinate frame of M . So $(U, (x^i))$ is an oriented local coordinate chart. Thus every point lies in an oriented local coordinate chart, and thus the orientation is continuous as desired.

- (5) Let $q: S^1 \times \mathbb{R} \rightarrow M$ be the quotient map, and let $\varphi \in \mathbb{Z}/2$ be the non-trivial element. Since q is a submersion between manifolds of the same dimension (since $\mathbb{Z}/2$ is discrete), dq_x is an isomorphism of tangent spaces for all x .

Suppose there is a global section $\sigma: M \rightarrow \text{Or}(TM)$. Since dq_x is an isomorphism, for each $(\theta, x) \in S^1 \times \mathbb{R}$ let $\hat{\sigma}(\theta, x) = dq_{\theta, x}^{-1}(\sigma[\theta, x])$. This maps a basis in $\sigma[\theta, x]$ to a basis of $T_{\theta, x} S^1 \times \mathbb{R}$, all the bases are consistently oriented. This is because essentially we are mapping a basis B to AB (where $A = dq^{-1}$); if C is another basis with X the transition matrix, so $XB = C$, then $Y = AXA^{-1}$ is the transition matrix from AC to AB , and $\det Y = \det X$.

Now $q \circ \varphi = q$ which means that

$$\hat{\sigma} = dq^{-1} \circ \sigma = d(q \circ \varphi)^{-1} \circ \sigma = d\varphi^{-1} \circ dq^{-1} \circ \sigma = d\varphi \circ dq^{-1} \circ \sigma.$$

But $d\varphi$ has a representation $\text{diag}(1, -1)$ which has determinant -1 . Thus $d\varphi$ reverses orientation, so $d\varphi^{-1} \circ dq^{-1} \circ \sigma$ would have orientation opposite to $dq^{-1} \circ \sigma$, a contradiction.

(6) Let us write

$$\text{Or}(E) = \{(p, \mathcal{O}_p) \mid p \in M, \mathcal{O}_p \text{ is an orientation of } E_p\}.$$

We have that

$$\pi^*E = \{(e, \mathcal{O}_p) \mid e \in E_p, \mathcal{O}_p \in \text{Or}(E)_p\},$$

The fibers of π^*E over $\text{Or}(E)$ are isomorphic to E_p . Thus

$$\text{Or}(\pi^*E) = \{(e, \mathcal{O}_p, \mathcal{O}'_p) \mid e \in E_p, \mathcal{O}_p \text{ and } \mathcal{O}'_p \text{ are orientations of } E_p\},$$

since $\pi^*E_{(e, \mathcal{O}_p)} \cong E_p$ so orientations of the fibers of π^*E are orientations of the fibers of E .

Now define $\sigma: \text{Or}(E) \rightarrow \text{Or}(\pi^*E)$ which maps $\mathcal{O}_p \in \text{Or}(E)_p$ (an orientation of E_p) to $(0_p, \mathcal{O}_p, \mathcal{O}_p)$, where $0_p \in E_p$ is the zero vector. This is a smooth section, meaning $\pi^*E \rightarrow \text{Or}(E)$ is orientable as desired.

Now we claim that π^*TM and $T\text{Or}(TM)$ are isomorphic bundles. Since π^*TM is orientable, this would mean $T\text{Or}(TM)$ is an orientable bundle and thus by (4), $\text{Or}(TM)$ is an orientable manifold. We have

$$\begin{aligned} \pi^*TM &= \{(v, \mathcal{O}_p) \mid v \in T_pM, \mathcal{O}_p \in \text{Or}(TM)_p\}, \\ T\text{Or}(TM) &= \{(\mathcal{O}_p, v) \mid \mathcal{O}_p \in \text{Or}(TM), v \in T_{\mathcal{O}_p}\text{Or}(TM)\}. \end{aligned}$$

Now, since $\pi: \text{Or}(TM) \rightarrow M$ is a two-fold cover, it is in particular a local diffeomorphism. Thus $d\pi_{\mathcal{O}_p}: T_{\mathcal{O}_p}\text{Or}(TM) \rightarrow T_pM$ is an isomorphism.

Consider the map

$$\theta: T\text{Or}(TM) \rightarrow TM \times \text{Or}(TM), \quad \theta(\mathcal{O}_p, v) = (d\pi_{\mathcal{O}_p}v, \mathcal{O}_p).$$

This is smooth since its first component is just the differential $d\pi$, and the second is just a projection. Furthermore it lands in $\pi^*TM = TM \times_M \text{Or}(TM)$ since both coordinates map to p by π_{TM} and π . Since $\pi^*TM \subseteq TM \times \text{Or}(TM)$ is embedded, we can restrict θ to a smooth map

$$\theta: T\text{Or}(TM) \rightarrow \pi^*TM.$$

It is bijective: if $\theta(\mathcal{O}_p, v) = \theta(\mathcal{O}_q, u)$ then $\mathcal{O}_p = \mathcal{O}_q$ and since $d\pi_{\mathcal{O}_p}$ is an isomorphism, $v = u$. And for $(w, \mathcal{O}_p) \in \pi^*TM$, $\theta(\mathcal{O}_p, d\pi_{\mathcal{O}_p}^{-1}w) = (w, \mathcal{O}_p)$.

It also clearly commutes with the projections $T\text{Or}(TM) \rightarrow \text{Or}(TM)$ and $\pi^*TM \rightarrow \text{Or}(TM)$, and on the fibers it is linear: for $\mathcal{O}_p \in \text{Or}(E)$ θ restricts to $d\pi_{\mathcal{O}_p}$ on the fiber of $\mathcal{O}_p$. Thus θ is a bundle morphism, and as it is bijective, it is a bundle isomorphism.

So $\pi^*TM \cong T\text{Or}(TM)$ and as the former is orientable, so too is the latter. Then by (4) $\text{Or}(TM)$ is orientable as desired.

12.5 Problem

Let M be a smooth manifold. Denote interior multiplication by $X \in \mathfrak{X}(M)$ by ι_X .

- (1) Prove **Cartan's formula**: for every $\omega \in \Omega^k(M)$ and $X \in \mathfrak{X}(M)$,

$$\mathcal{L}_X \omega = d\iota_X \omega + \iota_X d\omega.$$

- (2) Suppose M is oriented with Riemannian metric g . Give ∂M its induced orientation, and let $\nu_{\partial M} \rightarrow \partial M$ be the normal bundle of $\partial M \subseteq M$. Prove that there exists a unique section $N \in \Gamma(\nu_{\partial M})$ with norm 1 such that for every $p \in \partial M$, if $(e_1, \dots, e_{n-1})$ is an oriented basis of $T_p \partial M$, then $(N_p, e_1, \dots, e_{n-1})$ is an oriented basis for $T_p M$.
- (3) Let $\nu_g \in \Omega^n(M)$ be M 's volume form. Let

$$\nu_{\iota^* g} \in \Omega^{n-1}(\partial M)$$

be ∂M 's volume form with respect to the metric $\iota^* g$ induced from the inclusion $\iota: \partial M \hookrightarrow M$. Prove that

$$\nu_{\iota^* g} = \iota^*(\iota_N \nu_g).$$

- (4) Since $\nu_g \in \Omega^n(M)$, for every $f \in C^\infty(M)$ the integral $\int_M f \nu_g$ is defined. Prove the **divergence theorem**: if $X \in \mathfrak{X}(M)$ is compactly supported then

$$\int_M \operatorname{Div} X \nu_g = \int_{\partial M} \langle X, N \rangle_g \nu_{\iota^* g}.$$

- (1) For $Y_1, \dots, Y_k \in \mathfrak{X}(M)$ we know

$$\mathcal{L}_X \omega(Y_1, \dots, Y_n) = X(\omega(\bar{Y})) - \sum_i \omega(Y_1, \dots, [X, Y_i], \dots, Y_n).$$

Now we compute $d\iota_X \omega(\bar{Y})$ and $\iota_X d\omega(\bar{Y})$. For an increasing multi-index I , let us denote by $\bar{Y}_{-I}$ the tuple obtained from removing the indices I from $\bar{Y}$:

$$\bar{Y}_{-i_1, \dots, i_\ell} = (Y_1, \dots, \hat{Y}_{i_1}, \dots, \hat{Y}_{i_\ell}, \dots, Y_k).$$

From the previous homework,

$$\begin{aligned} d\iota_X \omega(\bar{Y}) &= \sum_i (-1)^{i-1} Y_i (\iota_X \omega(\bar{Y}_{-i})) + \sum_{i < j} (-1)^{i+j} \iota_X \omega([Y_i, Y_j], \bar{Y}_{-i,j}) \\ &= \sum_i (-1)^{i-1} Y_i (\omega(X, \bar{Y}_{-i})) + \sum_{i < j} (-1)^{i+j} \omega(X, [Y_i, Y_j], \bar{Y}_{-i,j}) \\ &= \sum_i (-1)^{i-1} Y_i (\omega(X, \bar{Y}_{-i})) - \sum_{i < j} (-1)^{i+j} \omega([Y_i, Y_j], X, \bar{Y}_{-i,j}). \end{aligned}$$

The last equality is due to the antisymmetry of ω .

For $\iota_X d\omega(\bar{Y}) = d\omega(X, \bar{Y})$, we consider the removal of the first index and all other indices independently. Notice that in the second sum in the exterior derivative (removing two indices), we consider two cases: when $i = 1$ and $1 < j \leq k+1$ (removing X and Y_{j-1} and multiplying by $(-1)^{1+j}$), and when $1 < i < j$ (removing Y_{i-1} and Y_{j-1} and multiplying by $(-1)^{i+j}$). By permuting the summation index, in the first case we sum over all $j \leq k$, and multiply the term by $(-1)^j$. In the second case, we sum over all $i < j \leq k$, and multiply the term by $(-1)^{(i+1)+(j+1)} = (-1)^{i+j}$. Thus,

$$d\omega(X, \bar{Y}) = X(\omega(\bar{Y})) + \sum_i (-1)^i Y_i (\omega(X, \bar{Y}_{-i})) + \sum_{i < j} (-1)^{i+j} \omega([Y_i, Y_j], X, \bar{Y}_{-i,j}) + \sum_j (-1)^j \omega([X, Y_j], \bar{Y}_{-j}).$$

By our previous computation, we see that this is equal to

$$d\omega(X, \bar{Y}) = X(\omega(\bar{Y})) + \sum_j (-1)^j \omega([X, Y_j], \bar{Y}_{-j}) - d\iota_X \omega(\bar{Y}).$$

Furthermore, since $\omega([X, Y_j], \bar{Y}_{-j}) = (-1)^{j-1} \omega(Y_1, \dots, [X, Y_j], \dots, Y_k)$ since we apply $j-1$ transpositions to move the first index to the j th, this is equal to

$$\iota_X d\omega(\bar{Y}) = X(\omega(\bar{Y})) - \sum_j \omega(Y_1, \dots, [X, Y_j], \dots, Y_k) - d\iota_X \omega(\bar{Y}) = \mathcal{L}_X \omega(\bar{Y}) - d\iota_X \omega(\bar{Y})$$

as desired.

- (2) Let $n = \dim M$. Then since $\dim \partial M = n-1$, the dimension of the fibers of $\nu_{\partial M}$ are all 1. So there are only two vectors in each fiber with norm 1, each with opposite sign. Since $v, e_1, \dots, e_{n-1}$ and $-v, e_1, \dots, e_{n-1}$ have opposite orientations, for each $p \in \partial M$ there is a unique choice for N_p .

Let $(U, (x^i))$ be a boundary chart for M . The induced orientation of ∂M is such that if $e_1, \dots, e_{n-1}$ is oriented for ∂M then $-\partial_n, e_1, \dots, e_{n-1}$ is oriented. Here we use the convention that Euclidean half-space is nonnegative in its last coordinate. For $p \in U \cap \partial M$, let v be the orthogonal projection of $-\partial_n|_p$ to $\nu_{\partial M}|_p$.

Since $\nu_{\partial M}$ is orthogonal to $e_1, \dots, e_{n-1}$, $v, e_1, \dots, e_{n-1}$ remains a basis. Furthermore, since this is a projection orthogonal to the e_i we have that $-\partial_n = v + b^i e_i$ for some coefficients b^i . Thus the transition matrix between $-\partial_n, e_1, \dots, e_{n-1}$ and $v, e_1, \dots, e_{n-1}$ is of the form

$$\begin{pmatrix} 1 & & & \\ | & 1 & & \\ * & & \ddots & \\ | & & & 1 \end{pmatrix},$$

which has determinant 1. Thus $-\partial_n, e_1, \dots, e_{n-1}$ and $v, e_1, \dots, e_{n-1}$ are consistently oriented, so $v, e_1, \dots, e_{n-1}$ is oriented for M . Thus N_p must be the normalization of v , i.e. $N_p = v/|v|_g$, because N_p is unique and this vector has the desired property (since v does, and multiplying by a positive constant does not affect orientation). So locally in U , N is the projection of $-\partial_n$ onto $\nu_{\partial M}$ and then normalization. Both of these preserve smoothness (since $\nu_{\partial M}$ is a vector bundle), so N is smooth in U .

By uniqueness, if V is any other boundary chart, the N we defined for U and the N we defined for V must coincide by uniqueness of N_p at all points p . Since N is smooth over all boundary charts, it is smooth over all ∂M . It has the desired property in each U , and thus over all of ∂M .

- (3) Let $e_1, \dots, e_{n-1}$ be an oriented orthonormal basis in $T_p \partial M$. Then

$$\iota^*(\iota_N \nu_g)(e_1, \dots, e_{n-1}) = \nu_g(N_p, e_1, \dots, e_{n-1}).$$

By the previous point, $N_p, e_1, \dots, e_{n-1}$ is an oriented orthonormal basis of $T_p M$. The characterizing property of ν_g is that it maps oriented orthonormal bases to 1, so $\iota^*(\iota_N \nu_g)(e_1, \dots, e_{n-1}) = 1$.

This is true for all oriented orthonormal bases, and thus by the characterizing property of the Riemannian volume form, $\iota^*(\iota_N \nu_g) = \nu_{\iota^*g}$ as desired.

- (4) Since X is compactly supported, so too is $\text{Div} X$. This is because if $X = f^i E_i$ in a geodesic frame centered at p then $\text{Div} X(p) = (E_i f^i)(p)$. So if $X(p) = 0$ then $f^i = 0$ and thus $\text{Div} X(p) = 0$, meaning $\text{supp Div} X \subseteq \text{supp} X$ which is compact. So integrating $\text{Div} X \nu_g$ makes sense.

Previously we showed that $\mathcal{L}_X \nu_g = (\text{Div} X) \nu_g$. By Cartan's formula,

$$(\text{Div} X) \nu_g = \mathcal{L}_X \nu_g = d\iota_X \nu_g + \iota_X d\nu_g.$$

Since ν_g is a top-form, its exterior derivative is zero (all $(n+1)$ -forms are zero) and thus we can simplify

$$(\operatorname{Div} X)\nu_g = d\iota_X \nu_g.$$

By Stokes' theorem,

$$\int_M (\operatorname{Div} X)\nu_g = \int_M d\iota_X \nu_g = \int_{\partial M} \iota^*(\iota_X \nu_g).$$

Now we claim that $\iota^*(\iota_X \nu_g) = \langle X, N \rangle \nu_{\iota^*g}$ on ∂M .

To show the equality of two top-forms, it is sufficient to show that at any point and for a single oriented basis they agree. This is because top-forms are determined by their value at any oriented basis (by multi-linearity and antisymmetry). So for $p \in \partial M$ let $e_1, \dots, e_{n-1}$ be an oriented orthonormal basis of $T_p \partial M$. Then $N, e_1, \dots, e_{n-1}$ is an oriented orthonormal basis of $T_p M$ and so $X = \langle X, N \rangle N + \sum_i \langle X, e_i \rangle e_i$ and thus

$$\begin{aligned} \iota^*(\iota_X \nu_g)(e_1, \dots, e_{n-1}) &= \nu_g(X, e_1, \dots, e_{n-1}) = \\ &= \langle X, N \rangle \nu_g(N, e_1, \dots, e_{n-1}) + \sum_i \langle X, e_i \rangle \nu_g(e_i, e_1, \dots, e_{n-1}) = \langle X, N \rangle \end{aligned}$$

where the last equality is because $N, e_1, \dots, e_{n-1}$ is an oriented orthonormal basis and thus, by the characterization of the Riemannian volume form, $\nu_g(N, e_1, \dots, e_{n-1}) = 1$; the other terms are all zero by antisymmetry.

On the other hand, since $\nu_{\iota^*g}(e_1, \dots, e_{n-1}) = 1$ we see that

$$\langle X, N \rangle \nu_{\iota^*g}(e_1, \dots, e_{n-1}) = \langle X, N \rangle = \iota^*(\iota_X \nu_g)(e_1, \dots, e_{n-1})$$

as desired.

All in all, we have shown $\iota^*(\iota_X \nu_g) = \langle X, N \rangle \nu_{\iota^*g}$ and thus

$$\int_M (\operatorname{Div} X)\nu_g = \int_{\partial M} \iota^*(\iota_X \nu_g) = \int_{\partial M} \langle X, N \rangle \nu_{\iota^*g}$$

as desired.